

Galileo

Galilean Relativity

How fast are you moving?

How Fast Are You Moving?	Relative to What
0 m/s	Your Chair
400 m/s	Earth Center (Rotation)
30 Km/s	The Sun (Orbit)
220 km/s	Galaxy Center
370 km/s	The CMB

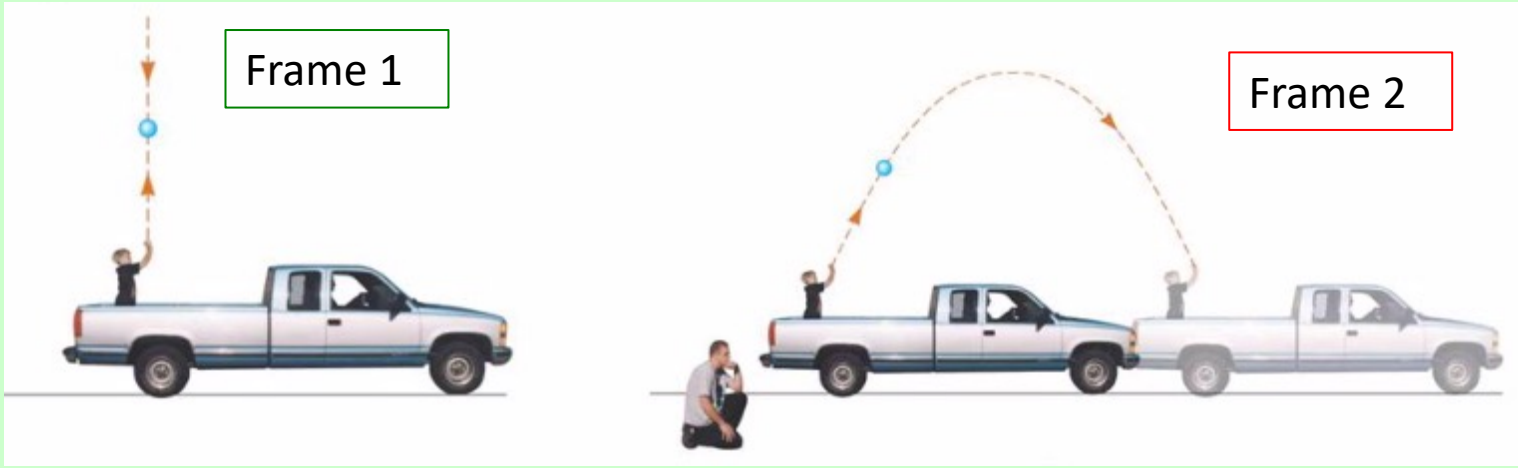
Relative to What??

This is part of the gist of special relativity

- It's the exploration of the physics of relative motion
- Only relative velocities matter: **no absolute frame**

A *Reference Fame* is a particular vantage point

1. What does the person in the truck bed see? (Frame 1)
2. What does the person outside the truck see? (Frame 2)



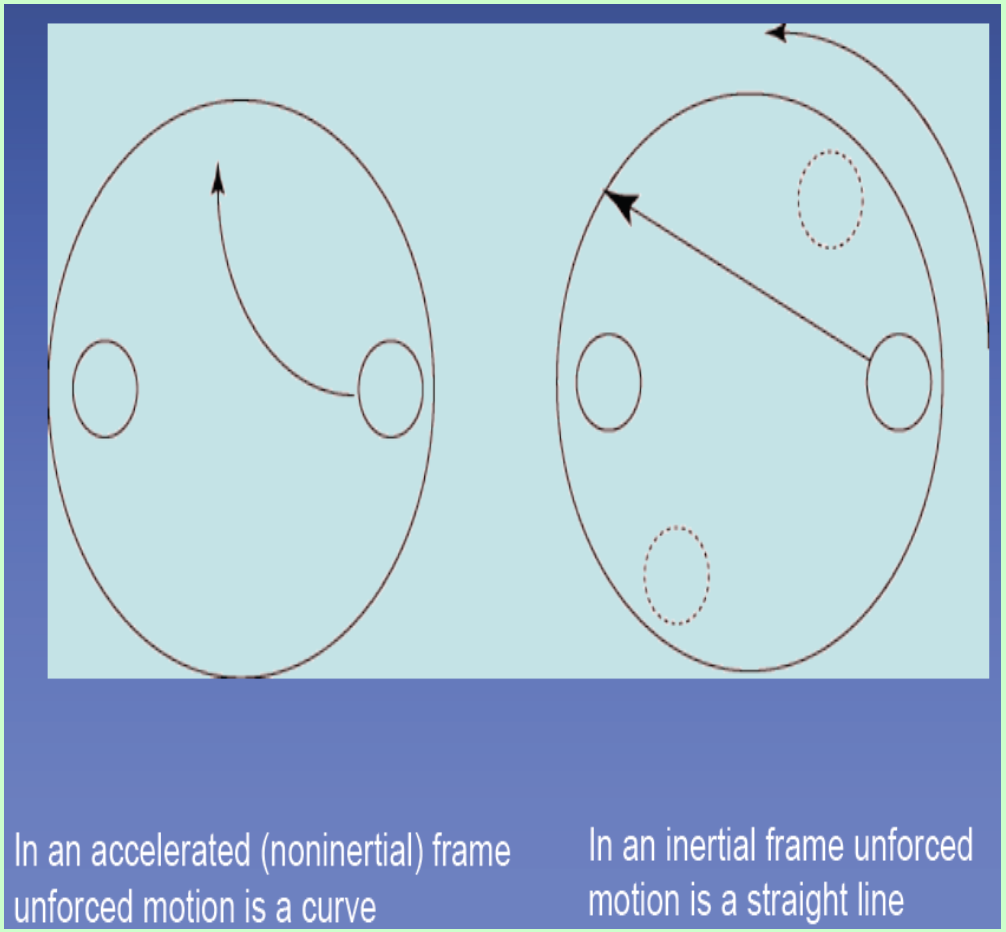
Relative: position, velocity (quantities tied to coordinate system)
Invariant: acceleration, mass, spacetime intervals, force (physical quantities)

Is this classroom standing still, or moving?

As long as its not accelerating, we have no way of knowing.
We feel no force!

1. An *Inertial Reference Frame* is one that is not accelerated
   – Objects move in straight lines with constant velocities

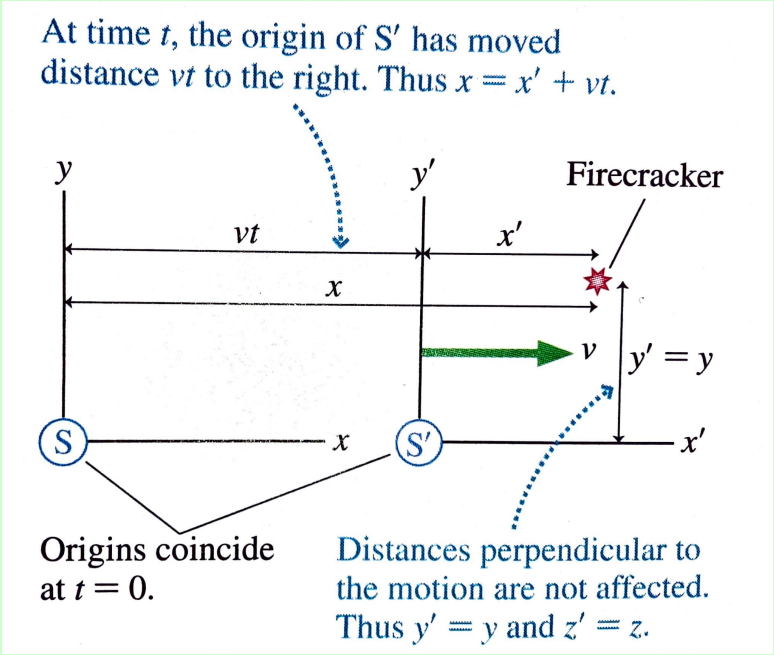
1. A non-inertial reference frame is one that is accelerated
   – Objects move in bent lines



Demo: non-inertial reference frame

Galiean Relativity A system of equations to relate two intertial frames, S and S′

Ex. A train is moving in the +x direction with velocity v. Reference frame S is the station outside, reference frame S′ is aboard the train. At time t a firecracker goes off. This is the EVENT E(t,x,y,z)

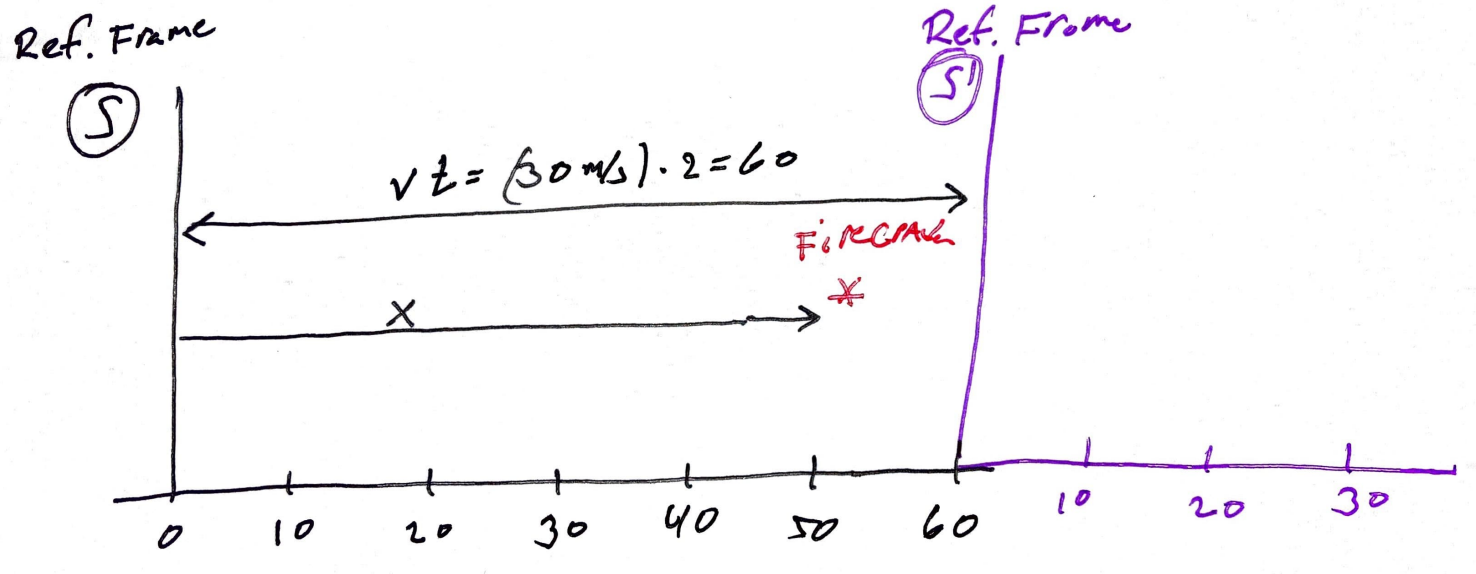


$$x = x' + vt \qquad x' = x - vt$$
$$y = y' \qquad \text{or} \qquad y' = y$$
$$z = z' \qquad z' = z$$
$$t = t' \qquad t' = t$$

Galilean Transformations

Whats the difference? We are assuming that we have all the info in S, and that S'
Is moving in the +x direction from S's perspective

Ex. A car is driving east at 30 m/s passes a boy at t = 0 seconds. The boy sets off a firecracker that goes off at x = 50 m at t = 2 seconds. Where does the drive say the firecracker goes off?



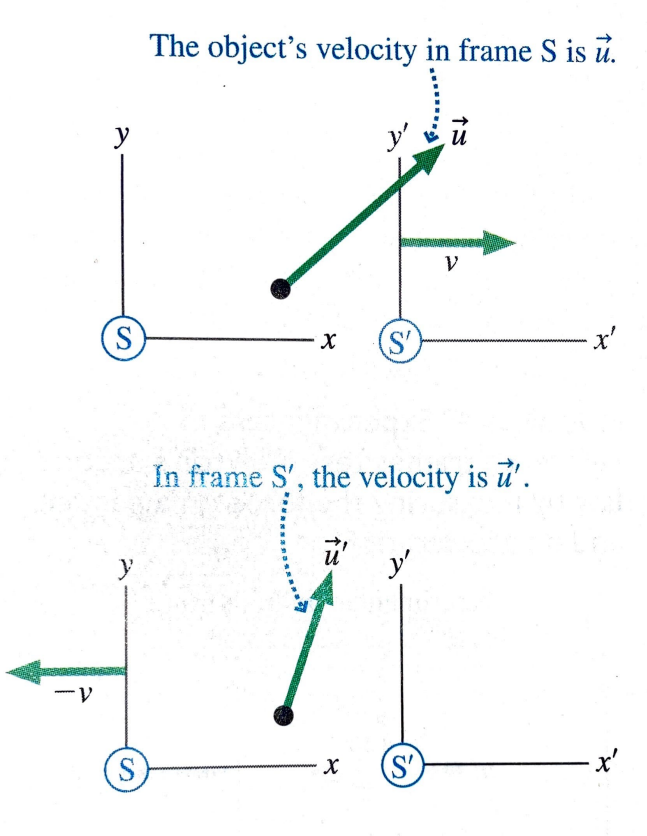
What are the coordinates of the Firecracker event in S' ?

$$x' = x - vt = 50 - 30(2) = -10 \text{ m}$$
$$y' = y = 0$$
$$z' = z = 0$$
$$t' = t = 2 \text{ s}$$

Can write the event Like:

E(t,x) = (2, 50)
E(t', x')= (2,-10)

What if an object is moving in a reference frame, but the frame is also moving relative to another frame?



Relativity: **u** is velocity in a frame
v is velocity of a frame rel to another

Here: **v** in +x, but **u** has components in x and y

$$u_x = \frac{dx}{dt} = \frac{d}{dt}\left(x' + vt\right)$$
$$= \frac{dx'}{dt} + \frac{d}{dt}(vt)$$

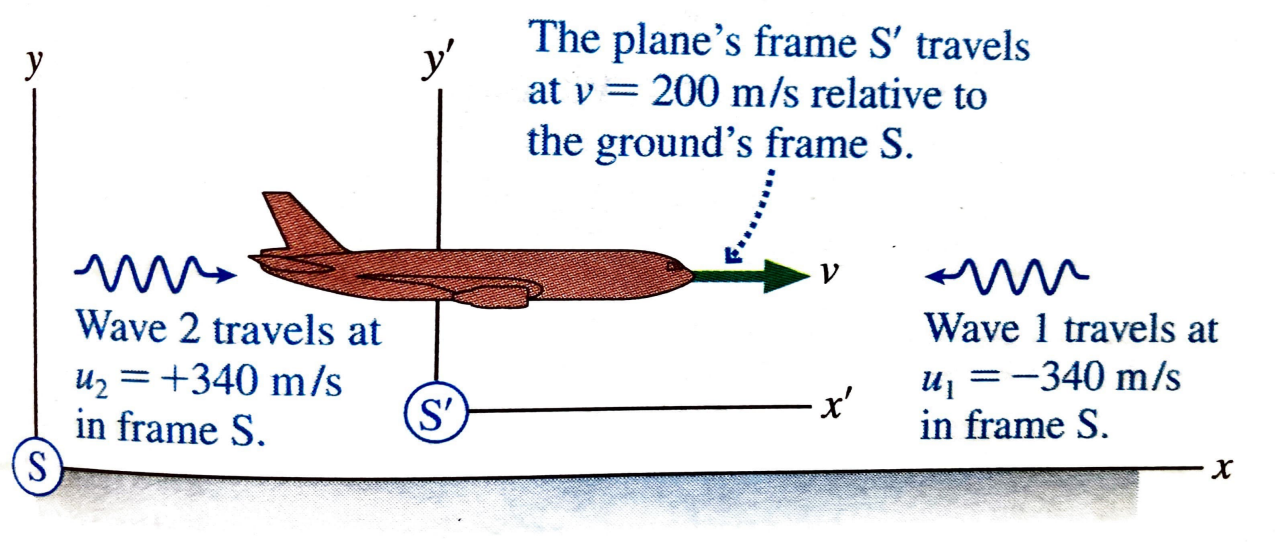
$$\boxed{u_x = u_x' + v} \quad \text{or} \quad \boxed{u_x' = u_x - v}$$

$$u_y = \frac{dy}{dt} = \frac{dy'}{dt} = u_y'$$

Full velocity transformation for S′ moving in +x direction

$$\boxed{\begin{aligned} u_x' &= u_x - v \\ u_y' &= u_y \\ u_z' &= u_z \end{aligned}}$$

Ex. Airplane is flying at speed 200 m/s w.r.t the ground. Sound wave 1 approaches plane from front, soundwave 2 is catching up from behind. Both travel at 340 m/s relative to the ground. What is the speed relative to the plane?



$$u_1' = u_1 - v = 340 - 200 = \boxed{140 \text{ m/s}}$$

$$u_2' = u_2 - v = -340 - 200 = \boxed{-540 \text{ m/s}}$$

Note: I find the signs in these equations confusing, but my intuition is never wrong like I knew u1 in S' should be 140 m/s. I usually use my intuition then randomly futz with the signs to make equations work using u' = u-v as my template.

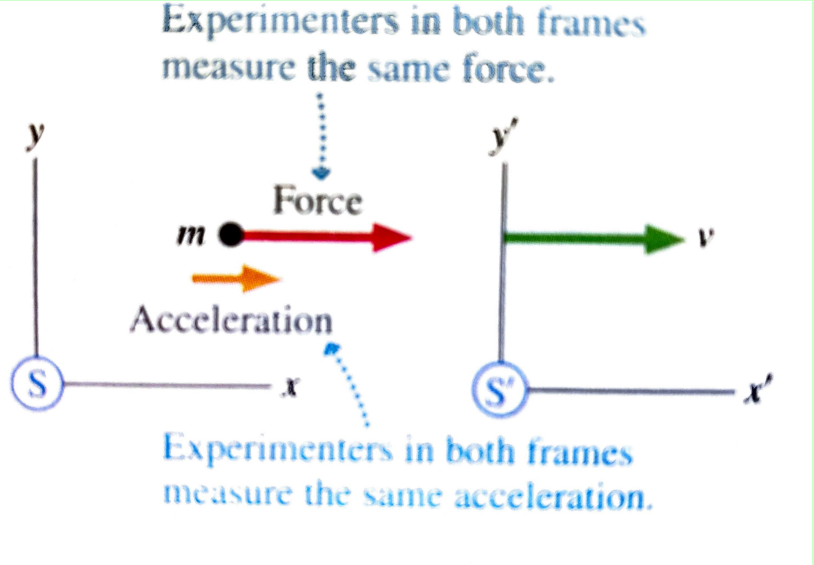
The Principle of Relativity

Suppose a force causes a ball to Accelerate in frame S. What does an observer in frame S' measure?

(assume v in +x for simplicity)

$$a' = \frac{du'}{dt} = \frac{d}{dt}(u - v) = \frac{du}{dt} = a$$

$$\boxed{a' = a}$$ (so long as the frame isn't accelerating)



We can extrapolate this to Newtons 2nd Law…, $\sum F = ma$, if its valid in one inertial frame (IRF), it must be valid in another.

The Galilean principle of relativity: The laws of mechanics are the same in all IRFs

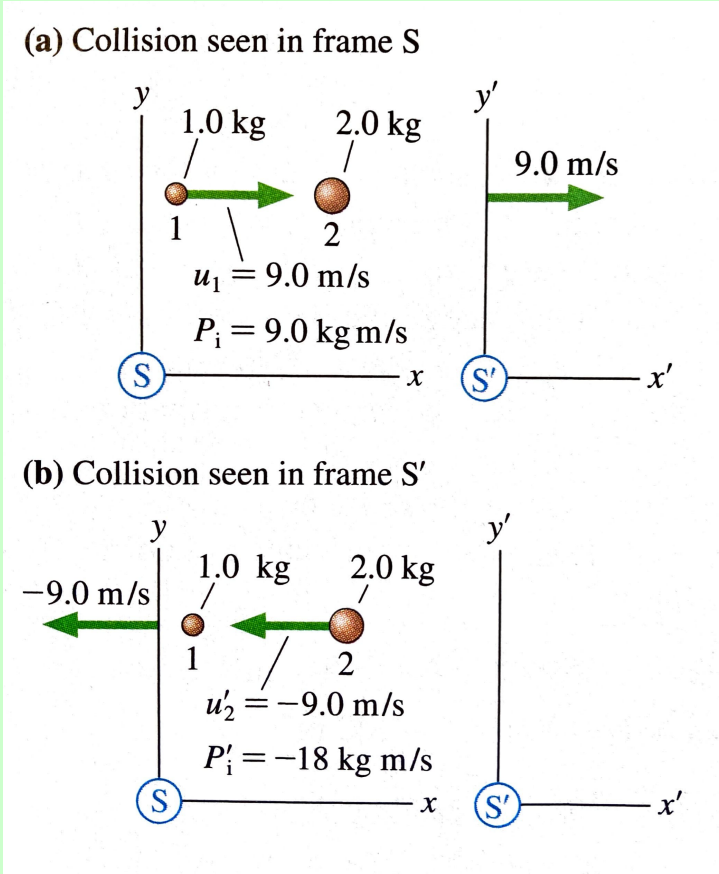
Galilean Relativity isn't saying that the value of all observables is the same in all IRFs. But rather that Newtons Laws and the conservation laws hold.

Ex. The Law of Conservation of Momentum

Particles 1 and 2 in S
$u_1 = 9$ m/s, $m_1 = 1$ kg
$u_2 = 0$ m/s, $m_2 = 2$ kg
S′ moves v =9 m/s

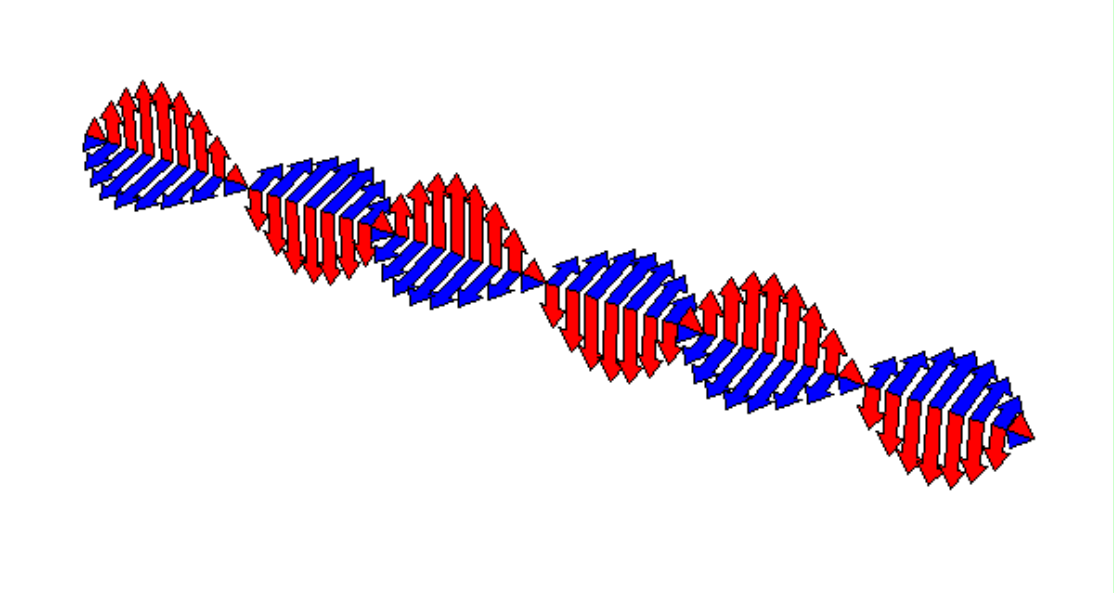
A. Find Pi and argue that Pi=Pf in S

B. Argue that Pi=PF in S'



Does light work like this?

What I see as light passes me:



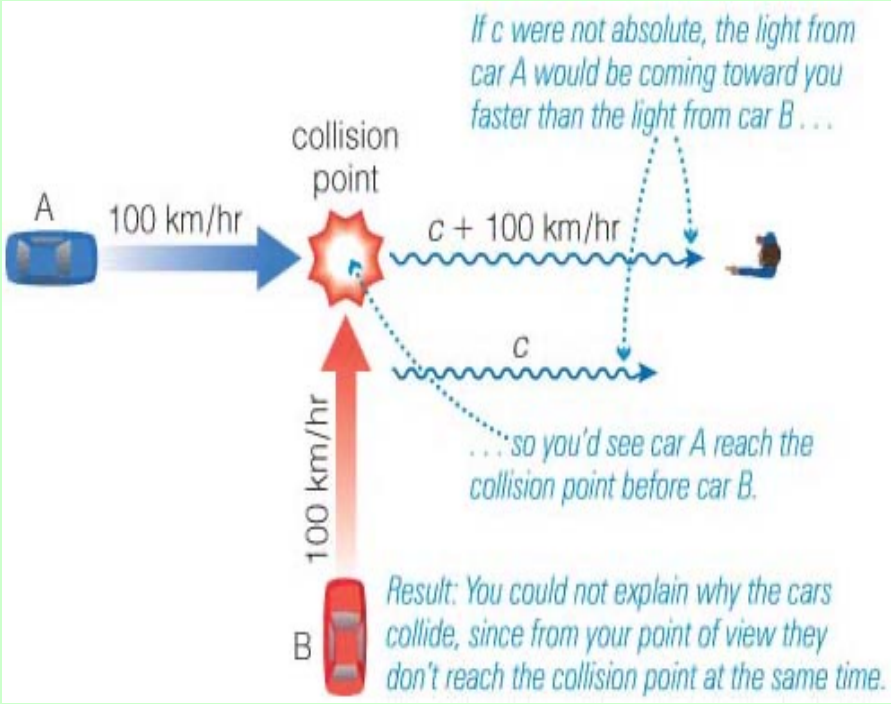
If I moved at the speed of light:

I wouldn't see any changing Electric and magnetic fields. The photon collapses!! Can't be! Laws of physics need to be the same in all IRFs. The speed of light must be the same for everybody

1. Light always travels at c

$$c = \frac{\text{Distance}}{\text{Time}} = \frac{d}{t}$$

2. Space and time as separate concepts must end



If *c* were not constant things would be still strange…

Newton: Space and time are absolute, c relative
Einstein: Space and time are relative, c absolute